

AiR: An Augmented Reality Application for Visualizing Air Pollution

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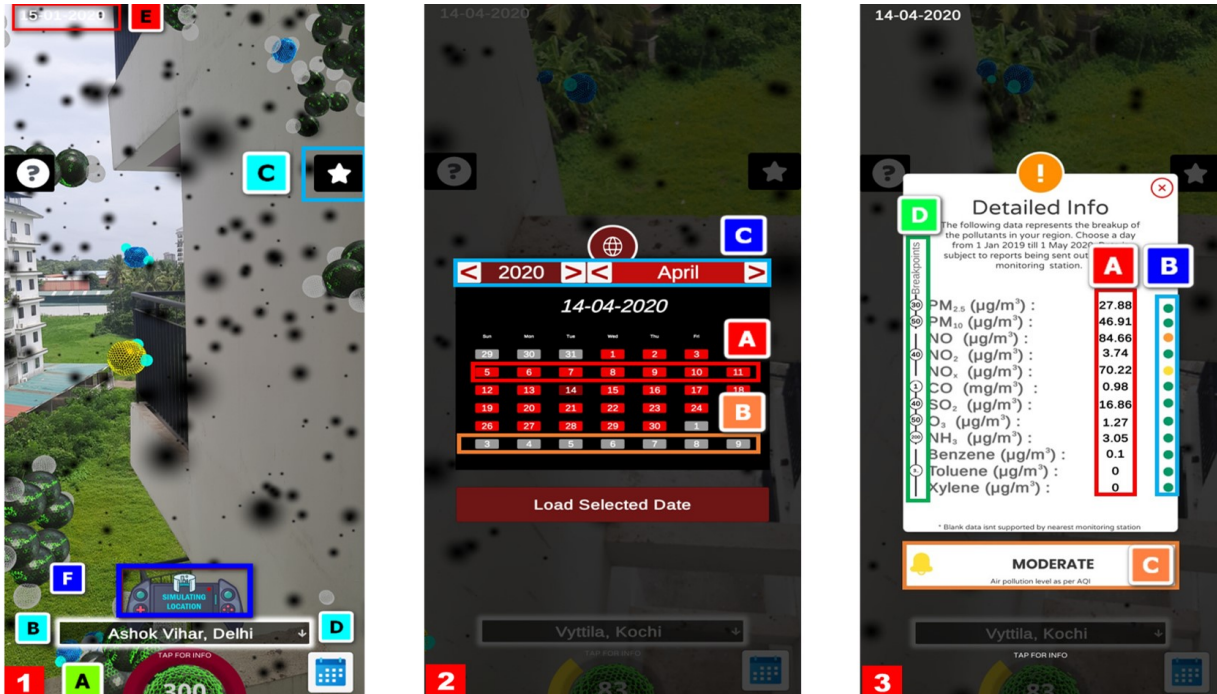


Figure 1: Snapshots of the Historical layout of *AiR*: (1) [A] AQI info-graphic [B] Location Dropdown [C] Live toggle [D] Date Selector [E] Selected Date [F] Location override (2) Date picker pop-up [A] Available Dates [B] Unselectable dates [C] Control toggles (3) Detailed info-panel [A] Pollutant Concentrations [B] AQI Indicator [C] Historical Air Quality [D] Breakpoints for good state

ABSTRACT

In order to effectively combat Air Pollution, it is necessary for the government and the community to work together. Easily comprehensible visualizations can play a major role in drawing public attention and spreading awareness about seemingly intangible air pollution. Considering the widespread usage of Android-based devices, in this paper, we propose an Augmented Reality based application called *AiR*, to help users to visualize pollutants in the air and to create an immersive user experience. It aims to interactively engage a wide variety of users and create awareness without overwhelming them with data. *AiR* visualizes 12 pollutants [PM_{10} , $PM_{2.5}$, NO, NO_2 , NO_x , CO, SO_2 , O_3 , NH_3 , C_6H_6 , $(CH_3)C_6H_5$ and $(CH_3)_2C_6H_5$] through unique models. We demonstrate our application on pollution data by *CPCB* from various weather stations across India collected over the initial lockdown period due to COVID-19 in India.

Index Terms: Human-centered computing—Visualization—Visualization systems and tools—; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Applied computing—Physical sciences and engineering—Earth and atmospheric sciences—;

1 INTRODUCTION

Air pollution has become a major cause of concern across the world with drastically increasing attention because of its impact on human health and welfare of the society. This has affected developing countries disproportionately due to rapid development and industrialization. In 2017, the share of deaths attributed to air pollution ranged from 2% in high-income countries, to nearly 15% in South and East Asian countries [1]. India is one such country which has seen many of its cities ranked in the top 10 most polluted cities in the world¹. Air pollution has been reported as the third-highest cause of premature deaths in India ranking just above smoking, and over 1.2 million Indians have lost their lives due to exposure to unsafe air in 2017 alone [25].

Frequent occurrences of thick haze in various parts of the country have become a major cause of concern over the years [27]. Despite considerable research and numerous policies² implemented by the government and various agencies like Central Pollution Control Board (*CPCB*), Ministry of Environment, Forest and Climate Change (MoEF&CC) in the country since 1981, the ground reality is that the improvements are slow and often lack support from the stakeholders which in turn leads to poor enforcement [29]. Most of

¹<https://www.iqair.com/world-most-polluted-cities/world-air-quality-report-2019-en.pdf>

²<https://cpcb.nic.in/air-pollution/>

the highly polluted cities in India have very high population densities as well. Life expectancy of an average person can be cut short by illnesses [2], which may be caused or worsened by the poor air quality in these cities. People often unaware endure high air pollution levels as a small price to pay for better livelihood options, which in the long term can lead to severe negative implications [14].

The issues in enforcement are commonly attributed to a failure in communication of the issues in a manner that is understandable to the general public [16]. Realizing this issue, the Government of India along with IIT Kanpur, released the National Air Quality Index (AQI)³ to strengthen air quality information dissemination system and in turn to increase awareness. AQI has become the most commonly used criterion in many countries including India, to assess the ambient air quality [3]. AQI has helped in standardizing and quantifying air pollution, but it is also important to make it easy to understand and to visualize the impact of pollution to the general public [11].

Researchers have proposed multiple solutions aimed at visualizing air quality data. Many of these approaches require users to interact with dedicated handheld devices or web portals [10, 11, 17]. Providing visualizations through mobile applications could further ease the task of reviewing air quality levels and help it grow into an integral part of the decision making process of the public and the government [11].

A common concern in most of the existing approaches so far has been the inability to present air pollution as something of immediate concern to someone who is not directly experiencing its impacts or is unaware of them. Essentially, people fail to act when they do not care or when they do not know the implications of a particular aspect on them [7]. A study by Anja Kollmuss and Julian Agyeman [18], highlights several factors that exist as barriers to pro-environmental behavior. Unlike a previous study by HJ Fietkau et al. [12], the authors do not directly link environmental knowledge to pro-environmental behavior. Though it certainly acts as one of the factors, there are many other key concerns that they bring up in their study, such as lack of internal incentives, socio-economic factors, and old behavior patterns [18].

Considering the extensive usage of the Android-based mobiles in India⁴, providing context specific air quality visualization on the Android platform could be helpful. Spreading awareness about different pollutants, their ill effects and sources in this manner could help inculcate pro-environmental behavior in the country and its youth [30]. However, to the best of our knowledge, we are not aware of any other work that visualizes multiple pollutant particles and allows visualizing trends over time and at multiple locations in an interactive manner through a mobile application and with the help of Augmented Reality (AR), motivating the need for our work. We present *AiR* as a way to visualize air pollutants using AR on the Android platform.

2 RELATED WORK

Technological advancements over the past decade have inspired many to develop visualizable air quality metrics based on national air quality standards. Some are aimed at presenting health impacts and others at representing air quality data to the general public in ways that are easy to understand. Many of these approaches were also designed to assist in data interpretation and pollution control measures [32].

A variety of attempts have been made in this field, many without relying on 3D visualizations. One such example is CAVisAP [24] for personalized outdoor air pollution visualization. The system provides visualizations that take into account location and time. Tzu-Chi Huang et al. came up with Smart-MapReduce [13], which aims

at helping in visualizing urban pollution using cloud computing technology. It tackled the problems with displaying vertical distribution of air pollution, which could not be done by common 2D air pollution maps. 3D volumetric clouds have been used to display air pollution concentrations at different levels from the ground aimed at modeling pollution dispersion [31]. The interface was also used to simulate impact that would be caused by a proposed urban transport scheme.

Huan Li et al. [19] used heat map based approaches to visualize and analyze variations of air pollutants in China to facilitate easy understanding of changes in air quality. Systems have also been designed towards assisting domain experts interpret patterns of Air Pollution through graph visualization [8] Muñoz-Pandiella et al. [23] describe visualization weathering effects in polluted cities to create an interactive and complex model taking into account the pollution distribution. Handheld devices to detect and generate personalized air quality have also been developed with an aim to increase public awareness on air quality [22].

Augmented Reality is being widely used in various domains over the past few years. AR takes advantage of virtual objects overlaid onto the surrounding real environment, resulting in a mixed reality where both coexist. AR has also been used to enhance learning of socio-scientific issues [4]. As pointed out in the study by Hsin-Yi Chang et al. [6], AR helps enhance one's sense of presence to invoke a feeling of immediacy and inspires conversations on the topic of concern. Recent examples of projects undertaken to visualize air pollution include New York times interactive website⁵ and "Aire", which allows users to learn about contaminants in the environment and helps visualize scientific concepts revolving around air pollution which was implemented as an iOS app [28]. Liu and Cranshaw [21] further explore how environmental data can be narrated for public awareness and to promote civic engagement.

However, most of the existing approaches in the literature, aimed at visualizing air quality are either observed to display complex visualizations or often have limited interactivity and lack engaging multi-perspective visualizations of air pollution analysis. Specifically, existing works in the literature do not leverage the potential of AR and are not available on the Android platform. These factors necessitate the development of solutions that could keep users well informed of the air quality, irrespective of their backgrounds. This could facilitate in bringing a sense of involvement and in turn help in reducing activities harmful to the environment by promoting civic participation.

In this paper, we present *AiR* as an Android based mobile application capable of representing air quality information for comparison using AR across different locations and varying dates. Using AR, we build a system that can help a user easily grasp the quantitative information about the pollutants around and also visualize it. The approach of *AiR* lets users explore a simulated virtual space and draw their own conclusions rather than overloading them with a lot of information. The open beta app uploaded on the Google Play Store⁶ was downloaded over 1000 times. The app has received a rating of 4.5 out of 5 from 37 reviewers and an overall positive feedback. The app is pending a formal evaluation, but we believe that this type of visualization is something which might appeal to the millennials.

3 DESIGN AND DEVELOPMENT OF *AiR*

In this section, we describe the design and steps taken towards the development of *AiR*.

⁵The NY website compares the users' city with the most polluted cities <https://www.nytimes.com/interactive/2019/12/02/climate/air-pollution-compared-ar-ny.html>

⁶<https://play.google.com/store/apps/details?id=com.iittp.air>

³<https://cpcb.nic.in/National-Air-Quality-Index>

⁴<https://gs.statcounter.com/os-market-share/mobile/india>

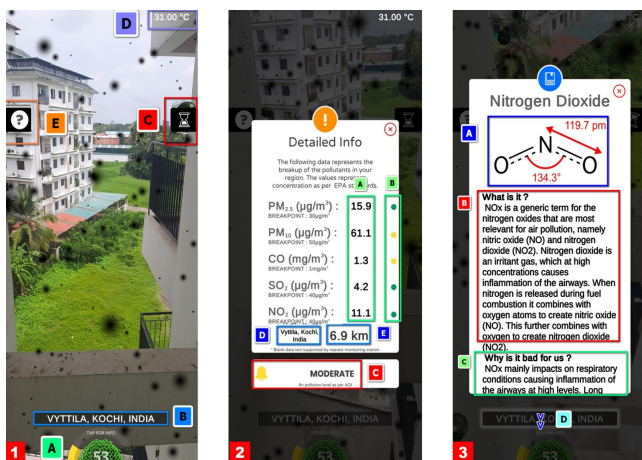


Figure 2: Snapshots of the Live layout: (1) Start screen: [A] AQI info-graphic button [B] Selected Location [C] Historical toggle [D] Temperature Info [E] About option (2) Detailed info panel: [A] Pollutant Concentrations [B] AQI indicator [C] Current AQI [D] Nearest station [E] Distance from station (3) Pollutant info panels: [A] Molecular Structure [B] Pollutant Info [C] Ill effects [D] Sources

3.1 Design Methodology and Goals

AiR : Air Pollution Visualizer has been developed to help immerse the user into an interactive AR simulated environment. It aims at overcoming knowledge barriers to pro-environmental behavior [18], thus effectively drawing the attention of citizens to air pollution. The underlying objective being, nudging users to make responsible decisions and keep the atmosphere clean. Being aware of the pollution levels in their surroundings and the ill effects might help people realize the urgency of taking pro-environmental decisions.

Hronn Brynjarsdottir et al. point out how such persuasive sustainability based arguments rely on the assumption of rationality [5]. In their work they suggest addressing this issue by allowing users to draw conclusions themselves to be more connected with the data and in turn induce behavioural change. With this in mind, we built *AiR* towards 3 core objectives which encompass the motivation behind it and define its design rationale.

- Provide a memorable interactive visualization [20] experience of air pollutants through Augmented Reality.
- Users should be able to perceive the relevant compartmentalized information, be able to draw conclusions by relating visually between depicted pollution levels at different dates and locations.
- Help provide a means to bridge the barriers to Pro-Environmental behavior faced by the people [18] by making the seemingly intangible feel tangible.

3.2 System Design and Implementation

When the *AiR* app is started, it queries the user's location automatically and gets the GPS coordinates of the user. With the GPS coordinates, the app estimates the distance to the weather stations and finds the closest active one using an optimized Haversine formula⁷. The latest air pollution data is retrieved from this station and used to create the visual representation of air quality. The user's camera feed is then augmented with 3D clickable objects which represents the various types of pollutants. It also has a dashboard to show a breakup of all the pollutants their concentrations and

⁷https://en.wikipedia.org/wiki/Haversine_formula

individual color scales based on severity for easy understanding. The data is analyzed using standards set by *CPCB*⁸ and is used to generate individual AQI values for the pollutants in the dashboard. The cumulative AQI score is displayed at the bottom of the screen with a color coded radial gauge for easy reference (Fig. 1(1)[A]).

The detailed info panel also includes a pollution level indicator (Fig. 2(2)[C]) to signify the meaning of the current AQI level to the end user as per the National Air Quality Index system adopted by India⁹. This view also shows the distance of the user from the closest weather station from which data has been sourced to account for ambient pollution metrics in use (Fig. 2(2)[E]).

On each query, an airspace of 10m x 10m x 5m (500m³) around the user is used to visualize air pollution. Number of particles of each pollutant is determined by checking its deviation from the mean. The application currently caps the maximum count of particles to prevent glitches on older devices. The pollutants are randomly distributed into this airspace based on the air pollution data, which allows the user to experience this simulation and visualize the perspective through their own device and move around instilling a broader sense of involvement. The live layout also retrieves and displays the current temperatures for an overall picture (Fig. 2(1)[D]). The view is refreshed if the user reaches the boundaries of this airspace.

All the pollutants have unique color coded 3D models to distinguish them in the simulated space. The molecules are built based on the chemical structures. The view is also built to be interactive, allowing the user to tap on any of the pollutants and learn about them. Discovering a pollutant and tapping on it triggers a pop-up containing the details about the pollutant, its sources and effects (Fig. 2(3)). Tapping on the hourglass icon positioned on the right hand side on the screen (Fig. 2(1)[C]), the user is taken to the historical layout shown in Fig. 1 which allows him/her to compare the data through AR from multiple weather stations from all over the country on various dates.

Once the nearest location is queried the date-picker on the bottom right (Fig. 1(1)[B]) is updated to show the available dates for the selected station. If the user selects a location other than the closest weather station, a graphic overlay alerts the user that the location is being simulated to avoid confusion (Fig. 1(1)[F]).

Currently, there are 12 unique pollutants within the app [PM₁₀, PM_{2.5}, NO, NO₂, NO_x, CO, SO₂, O₃, NH₃, C₆H₆, (CH₃)₂C₆H₅ and (CH₃)₂C₆H₅]. All of them have unique information panels within the app (Fig. 2(3)).

3.3 Development

AiR provides an easy to comprehend overview of the air quality in the user's locality and also provides an option to switch location and dates over the past year. The app was built with the help of the AR Foundation framework, which helps leverage the power of ARCore in Android¹⁰. The workflow utilized to develop the app is shown in Fig. 3. The first task was to collect data, for which we processed air pollution data since 2019 in order to enable comparison with pre-COVID-19 levels.

Over the course of the project, daily data for the individual pollutants was appended to our database. All the raw data used in this project was sourced from the *CPCB* online portal. This portal provides us access to all the weather stations under the organization and the various state-level control boards. Daily data from January 2019 till first of July 2020 data was drawn and cleaned. The dates with missing data were removed if they extended for periods longer than 2-3 days after confirming unavailability. Other missing data points were interpolated in order to avoid glitches while working with the data. Models for each of the pollutants were built as mesh

⁸<https://app.cpcbcr.com/AQI/>

⁹<https://pcb.nic.in/naqi/>

¹⁰Supported Devices: <https://developers.google.com/ar/discover/supported-devices>

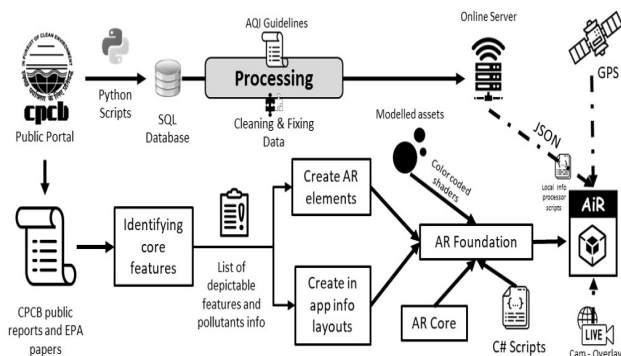


Figure 3: Workflow of building *AiR*

groups within Unity¹¹ and shaders were applied to them to optimize performance. Each of the individual elements of the molecules were color coded and geometrically placed as per their chemical structure for consistency throughout the app.

The quantitative control and spawning of the pollutants in the simulated airspace around the user was handled with C# Scripts. AQI and subindices (for each pollutant) were evaluated by building upon the standard system used by CPCB. Due to prominent pollutants varying in different regions and disproportional distributions, the max counts of each pollutant was defined after manual performance tests on a wide range of devices sourced from volunteers.

The particles are spawned at random locations and are imparted small velocities and rotations so as to represent air as the fluid it is. Particulate matter which has no specific size, but rather a range spawns at varied sizes to represent the same. However, this could be updated in future versions to make use of Junge [15] or other similar distributions. User taps are also registered and ray-casts are issued to identify particle being invoked. Data is displayed to the user for the corresponding particles with info about what it is, its molecular structure, how it is formed, and the ill effects it causes. This data is sourced from authorized Environmental Protection Agencies and public reports and warnings issued by CPCB. The fully processed data was supplied through a REST API as JSON responses for each location to minimize space requirements and background data used. On fresh launch, the app queries the server for historical data of the nearest weather station and the live data for the initial layout. On overriding the location within the historical layout, the request is sent again for the new location. The range of the camera is also set to three meters based on performance tests and to encourage the user to move around and discover the particles to help avoid a passive experience. The background scripts also keep track of the user's position from world origin to prevent going out of the simulated region.

4 DISCUSSION AND FUTURE WORK

With the advent of the internet and increasing affordability of devices to access it, the data which is available to a user is increasing day by day which may lead to information overload and at times may lead to important information getting ignored [26]. Several visual methods have been employed in bringing awareness in multiple social and educational domains. *AiR* is our attempt towards visualization of air pollution in an interactive manner, making it a unique learning experience. It allows the user to gauge the Air Quality level visually without the need to comprehend numerical or graphical data. This can be seen in Figure 4, here we see the rise in pollution levels following the easing of the Covid-19 Lockdown in the city of Delhi.

¹¹<https://unity.com>

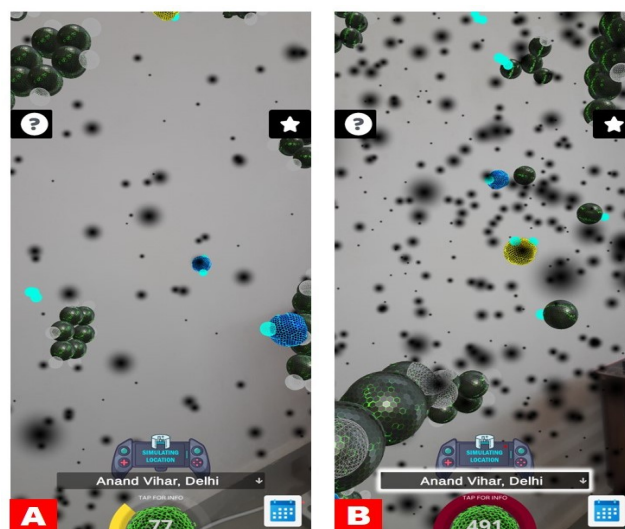


Figure 4: Subfigures A and B depict the city of Delhi during Unlock 3.0 and 6.0 phases respectively as seen within *AiR*

The AQI represents the Outdoor Air Quality or Ambient Air Quality. This, however, should not be mistaken for Indoor Air Quality as pointed out within the app as well. If a Continuous Air Quality Monitoring Station is near to the user it can influence the indoor air quality. Real world air pollution is a spatiotemporal event and hence is difficult to model accurately [9], however we attempt to provide a simplified approach to the visualization that can be easily understood and might be helpful to encourage adoption of environment-friendly practices.

In terms of the visualization, we plan on tweaking the parameters taken into account for the relative analysis to a more dynamic model. The AR experience can be further improved by adding depth map support and in turn improved occlusion. Also currently, not all weather stations are available for public view due to inconsistency in the data introduced by stations with erratic data and those which are no longer active. Furthermore, the database has processed data only from January 2019 until July 2020.

The application and methodology could be further extended and future research would be required to study the impact of deploying our solution in other communities with varied characteristics. Further user studies could also help look into the conclusions drawn by the users on using *AiR* when compared to the other available means of visualization. We also plan on mapping several other related parameters that could be correlated and consequently observe the inherent patterns among them which may otherwise not be obvious.

5 CONCLUSION

In this paper, we have discussed the need for bringing awareness among the public regarding air quality in reducing air pollution levels. The exponential increase in the use of Android mobile devices every month in India, continued improvements in AR technology and the increasing number of supported devices motivate its use towards *AiR* Pollution Visualization. Hence, we proposed *AiR* as an Android based mobile application that is capable of visualizing and comparing air quality across different locations and dates through AR. Unlike methods and visualizations currently available to the general public, *AiR* aims to be much more user friendly and appealing. Demonstration of the application on data before and after lockdown also reveals a concerning steep rise in air pollution levels in key cities across the country after the initial dip taking it back towards toxic levels.

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